

Identifying Underserved Cuisine Markets Using Big Data Techniques

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Abstract—Restaurant success often depends on choosing a good location, but many decisions are still made without strong evidence [1]. In this project, we created a system that uses Apache Spark to find areas in the United States where certain cuisines might be underserved [2]. We combined restaurant data from OpenStreetMap with demographic information from the United States Census and built one dataset at the zip code level [3], [4]. Our method uses two steps. First, we grouped similar markets using K-Means clustering [5]. Second, we trained a Random Forest model to estimate how many restaurants of a certain cuisine we would normally expect in each area [6]. By comparing the expected supply with the actual supply, we identified possible gaps for Pizza, Mexican, Coffee, and Chinese cuisines. The results show that large-scale data tools can reveal patterns that are not obvious when looking at basic demographics alone [1].

Index Terms—Big Data, Apache Spark, Machine Learning, K-Means, Random Forest, Site Selection.

I. INTRODUCTION

Choosing a location is one of the most important decisions for a restaurant owner. A neighborhood might have a strong population, good income levels, or a balanced age range, but it might already have many restaurants that compete for the same customers [6]. Another neighborhood might look quiet at first but may actually have real demand with very few options. Because of this, owners benefit from a way to measure both the people living in an area and the restaurant supply that already exists.

This project focuses on finding where certain cuisines might be underserved at the county level. To do this, we built a system that brings together large and different types of data, prepares them for computation, and models the connection between community characteristics and restaurant supply using Apache Spark [1].

Our main contributions are:

- 1) A Spark pipeline for combining geospatial data from OpenStreetMap and economic data from the census [1], [2], [5].
- 2) A K-Means model that groups similar types of markets into clusters [3].
- 3) A Random Forest model that estimates expected restaurant supply and produces a gap score for each cuisine [4].

II. BACKGROUND / LITERATURE REVIEW

Big Data methods are becoming more common in retail location planning and urban analysis [1]. Older location models often assume that customers go to nearby or larger shopping areas, which are sometimes called gravity models [6]. These older methods do not consider detailed variables such as income inequality, age distribution, or obesity rates, which all influence food choices.

A. Geospatial Data

OpenStreetMap is widely used for mapping because it covers a large amount of data and is updated often [2]. Since it is crowdsourced, it may have missing or incomplete entries, but it becomes useful when combined with structured datasets such as census information [5]. Earlier work on large geospatial data sets often used Hadoop-based systems, but Spark has become more popular because it supports in-memory computing and performs better on iterative machine learning tasks [1].

B. Modeling Supply

Many studies that use machine learning for retail focus on predicting sales or store performance. Our goal is different. We aim to estimate how many restaurants of a certain cuisine should normally exist in a given area if it behaves like other similar markets. This idea is related to work on sales potential modeling, where the goal is to set a baseline expectation for inventory or store count [4]. To do this correctly, we first group markets by similarity using clustering, then predict expected supply inside each group. This helps avoid direct comparisons between very different places, such as rural areas and major cities [3], [4].

III. PROPOSED METHODOLOGY

Our pipeline has three main stages. We combine the data, process the features, and then train the models within the Apache Spark environment [1].

A. Combining the Data

Restaurant data from OpenStreetMap uses zip codes, while census and food environment data are reported at the county level [2], [5]. Because zip codes and counties do not always align, we used the HUD zip-to-county crosswalk file. This file provides a residential ratio that shows how much of each zip

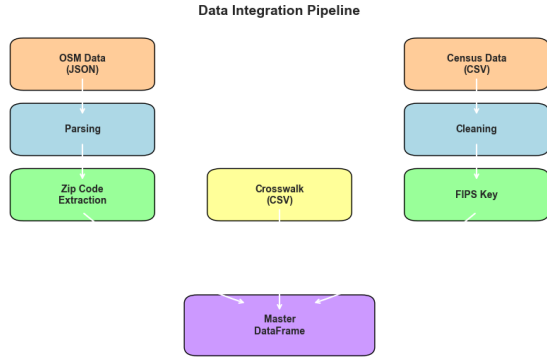


Fig. 1. Data Integration Pipeline.

code falls inside each county. We used these ratios to distribute county-level statistics across the zip codes, which allowed us to assign census values more accurately [1].

B. Processing the Features

Several economic features, such as Median Household Income, were heavily right-skewed in the raw data. We applied log transformations to reduce this skew and to improve the stability of the models [1]. After creating all numerical features, including income, poverty, obesity, and population density, we standardized them so they would have zero mean and unit variance. This step ensured that no single feature dominated the clustering or regression algorithms.

C. Modeling Steps

There are two main modeling steps in our pipeline.

1) *K-Means Clustering*: We used K-Means clustering to group counties into different types of markets based on their demographic profiles [3]. This step helps separate urban, suburban, and rural areas so that they are compared only to similar peers.

2) *Random Forest Regression*: Inside each cluster, we trained a Random Forest regression model to estimate the expected number of restaurants of a given cuisine per ten thousand residents [4]. Random Forest was chosen because it handles nonlinear relationships and interactions between demographic variables without strict assumptions on feature distributions [4].

D. Gap Score

After training the Random Forest model, we computed a gap score for each county and each cuisine:

$$\text{Gap} = \text{PredictedSupply} - \text{ActualSupply} \quad (1)$$

A positive value suggests that the area may be underserved, while a negative value suggests saturation.

E. Assumptions and Key Claims

To define the scope of our analysis, this project operates under several methodological and data-driven assumptions.

1) Methodological & Data Assumptions:

- **Proxy Reliability**: We assume that OpenStreetMap (OSM) data, while crowdsourced and potentially containing gaps, provides a statistically significant proxy for the actual supply of restaurants when aggregated at the zip code and county levels.
- **Demographic Determinism**: We assume that community characteristics—specifically income, poverty rates, obesity rates, and age structure—are valid predictors of restaurant demand.
- **Gap Interpretation**: We assume that a positive difference between our model’s predicted supply and the actual supply (a positive “Gap Score”) represents a market opportunity or “underserved” status.
- **Homogeneity within Clusters**: By using K-Means clustering ($k = 4$), we assume that counties assigned to the same cluster share enough economic and spatial similarities to serve as valid benchmarks for one another.

2) Project Claims:

- **Beyond Demographics**: We claim that standard demographic analysis is insufficient for site selection because it fails to account for existing saturation.
- **Predictive Validity**: The identification of high-gap counties supports the claim that Machine Learning can reveal “hidden” opportunities in suburban or intermediate markets.

IV. EXPERIMENTAL SET-UP

A. Datasets

We used three primary datasets in our analysis.

- 1) **OpenStreetMap (OSM) Data**: A 2.2 gigabyte JSON file containing raw geospatial points of interest. We filtered entries tagged as restaurant, fast food, or cafe [2]. This produced around 476 thousand food-related points.
- 2) **United States Census and Food Environment Data**: A structured table with county-level statistics including Median Household Income, poverty rate, obesity rate, age demographics, and population [5].
- 3) **HUD Zip to County Crosswalk**: A file that maps zip codes to county codes and provides a residential ratio for each pair [1].

B. Tools and Environment

We used Apache Spark 4.0 with PySpark for all parts of the pipeline [1]. Vector Assembler and the Spark ML library handled feature construction, K-Means clustering, and Random Forest regression.

C. Validation

We used several strategies to validate the models.

- **Clustering**: We tested several values of k and used the Silhouette score to select the number of clusters that gave good separation and cohesion [3].

- **Regression:** We used a 70-30 train-test split to evaluate the Random Forest model and checked that it generalized across different clusters.

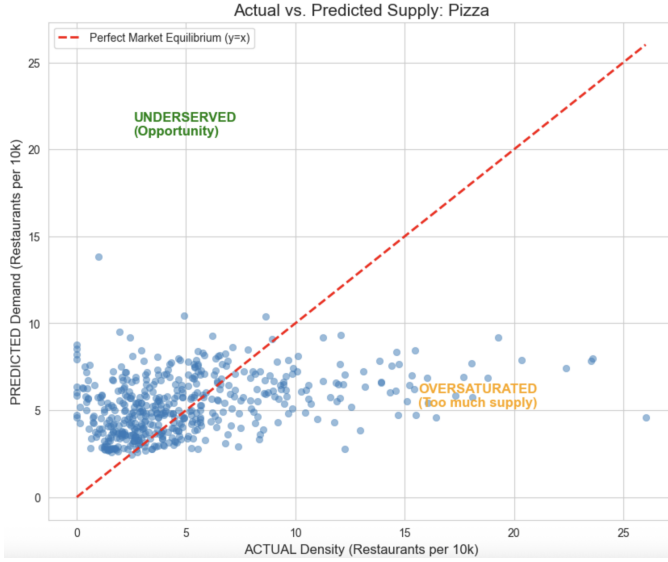


Fig. 2. Actual vs. Predicted Supply: Pizza. The model follows the equilibrium line ($y = x$).

As shown in Fig. 2, the model follows the diagonal equilibrium line ($y = x$) for the majority of counties, indicating that our predictions strongly correlate with reality.

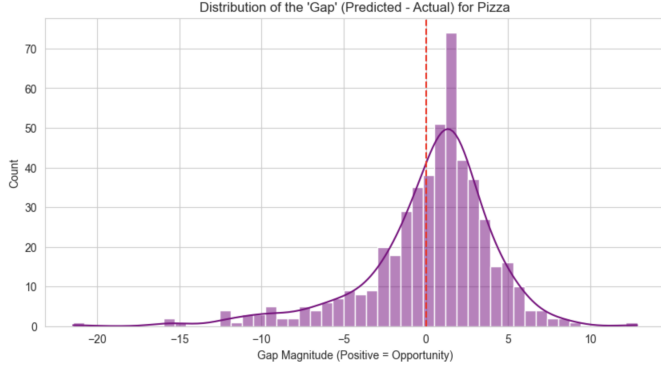


Fig. 3. Distribution of the 'Gap' (Predicted - Actual) for Pizza.

Gap Analysis: There is no official label for “opportunity,” so we manually reviewed the counties with the highest positive gap values. The distribution of the ‘Gap’ figure (Fig. 3) confirms that these deviations follow a normal distribution centered at zero, suggesting that large positive gaps are statistically significant anomalies.

V. RESULTS AND DISCUSSION

A. Initial Patterns

Exploratory analysis showed that county-level Median Household Income is not evenly spread out (Fig. 4). Most counties fall in the middle-income range, while only a smaller number have very high incomes.

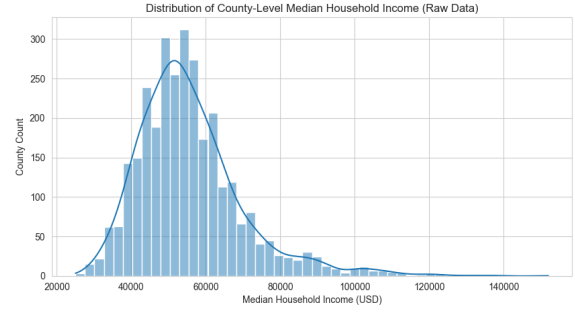


Fig. 4. Distribution of County-Level Median Household Income.

We also examined the distribution of restaurant types. A bar chart of average county-level counts showed that categories like Fast Food, Pizza, and American restaurants are very common, while others appear less frequently.

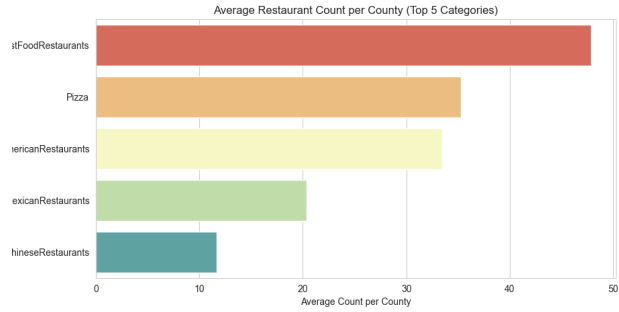


Fig. 5. Average Restaurant Count per County (Top 5 Categories).

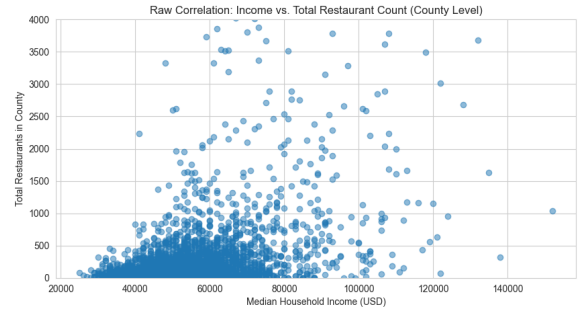


Fig. 6. Raw Correlation: Income vs. Total Restaurant Count.

Finally, a scatter plot comparing Median Household Income with Total Restaurant Count (Fig. 6) showed a positive correlation: counties with higher income generally tend to have more restaurants.

B. Clustering Results

We tested multiple values of k from 2 to 10 and found that $k = 4$ produced the clearest and most interpretable cluster structure [3].

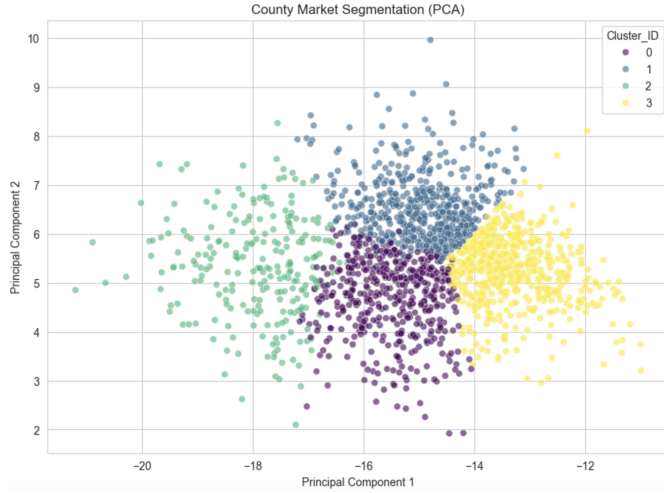


Fig. 7. County Market Segmentation (PCA).

When projected into two dimensions using PCA (Fig. 7), the groups corresponded to:

- **Cluster 0:** Middle-High Income / Rural-ish (e.g., Uinta, WY).
- **Cluster 1:** Middle Income / Rural (e.g., Taylor, WI).
- **Cluster 2:** High Income / Wealthy / Tourism (e.g., Teton, WY).
- **Cluster 3:** Lower Income / Working Class (e.g., Tyler, TX).

C. Cuisine Gap Analysis

The Random Forest regression was run separately for Pizza, Mexican, Coffee, and Chinese cuisines.

1) *Case Study: Pizza:* The model identified Tooele County in Utah (Cluster 0) as having the highest opportunity gap. With a Median Income of \$80,000 and an actual pizza density of 0.30, the model predicted a capacity for 6.06 pizza places per 10k residents.

TABLE I
PIZZA RECOMMENDATIONS BY MARKET TYPE

Cluster	Rank	County	Inc (\$)	Act	Gap
0	1	Tooele, UT	80k	0.30	5.77
0	2	Box Elder, UT	69k	1.67	4.45
1	1	Lancaster, SC	64k	0.43	4.89
1	2	Chelan, WA	60k	2.10	3.67
2	1	Calvert, MD	111k	0.44	5.40
2	2	Sarpy, NE	84k	1.43	2.96
3	1	Valencia, NM	55k	0.13	4.88
3	2	Coos, OR	50k	1.10	4.24

2) *Case Study: Chinese Cuisine:* Wise County, TX was identified as a prime location for Chinese cuisine. Despite a solid higher middle-class income profile (\$70,000), the area lacks Chinese restaurant options, generating a gap score of 1.89.

TABLE II
CHINESE REST. RECOMMENDATIONS BY MARKET TYPE

Cluster	Rank	County	Inc (\$)	Act	Gap
0	1	Wise, TX	70k	0.91	1.89
0	2	Box Elder, UT	69k	0.56	1.44
1	1	Lancaster, SC	64k	0.22	0.95
1	2	Yavapai, AZ	54k	0.13	0.90
2	1	Sarpy, NE	84k	0.22	1.37
2	2	Parker, TX	83k	0.37	0.98
3	1	Henderson, TX	50k	0.12	1.15
3	2	St. Mary, LA	41k	0.20	1.15

D. Discussion

The findings show that income alone does not determine opportunity. Several high-income urban cores had negative gap scores, pointing to saturation. In contrast, many suburban markets in intermediate clusters had solid purchasing power and relatively low supply.

The “Actual vs. Predicted Supply” analysis showed a clear positive correlation between the demographic capacity and restaurant inventory. However, the utility of this project lies in the deviations. The data points that fall significantly above the equilibrium line represent the “Gap,” confirming our assumption that specific markets are operating below their theoretical capacity.

This analysis fulfills the project goal of utilizing Big Data to reveal patterns not obvious through standard observation. For example, the identification of Tooele County, UT and Wise County, TX validates the claim that machine learning can pinpoint specific underserved neighborhoods that possess strong economic fundamentals but lack specific cuisine inventory.

VI. CONCLUSIONS

A. Summary

We created a Big Data pipeline for analyzing restaurant markets across the United States using Apache Spark [1]. By combining K-Means clustering with Random Forest regression, we developed a method to estimate where certain cuisines might be underserved. This approach moves beyond simple population maps and offers a more detailed way to measure market potential.

B. Limitations

This project has several limitations:

- OpenStreetMap data is user-generated and may miss restaurants or have incomplete tags [2].
- Zip codes do not perfectly match commercial boundaries.
- Cultural preferences and regional tastes are not directly modeled [6].

C. Future Work

Future work could improve this approach by incorporating review data (Yelp/Google) to weight restaurants by quality, analyzing time series trends to measure volatility, and adding mobility data to capture foot traffic.

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